

Cosmo-Edu Activities

Eleonora Panini^{1, 2}, and ...³

¹Department of Physics, FIM Faculty of Physics, Informatics and Mathematics,
UNIMORE Modena and Reggio Emilia University, Modena, Italy.

*Address correspondence to: eleonora.panini@unimore.it

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1 Dark Matter

- **Target topics:** Dark Matter (galaxy rotation curves, clusters).
- **Learning objectives:** Analyze multiple lines of evidence for dark matter; derive galaxy rotational velocity curve from mechanics, Newton gravitation law and Kepler law; investigate galaxies and clusters composition; find the cluster mass from virial theorem reviewing kinetic energy and gravitational potential; connect classical Physics to Cosmology.

- **Physics program link:** Mechanics, Kepler's law, Newton's law of gravitation, centripetal acceleration, circular motion, centripetal force, gravitational force, energy and momentum conservation, gravitational potential energy, kinetic energy.
- **Misconceptions:** "An object beyond the Earth's atmosphere (in space) does not experience a force of gravity.", "Moving an object higher from the Earth's surface increases the force of gravity."
- **Methodology:** Problem solving, data-driven, ISLE, cooperative learning, brainstorming, sonification, multisensing
- **Introduction**
 - Interactive brainstorming session (using LIM): Discuss cosmology, moving from Newton's and Kepler's laws to galaxies. View real graphs and data, and historical introductions to observations (e.g., Rubin, Zwicky). View introductory simulations and videos.
- **Prerequisite Check**
 - Quiz with multiple-choice or true/false questions on Newton, Kepler, and centripetal force. Goal: Verify prerequisites and identify any misconceptions.

1.1 Galaxy rotational velocity curve

- **Objectives:**
 1. Apply the law of gravitation to derive the expected orbital velocity in a galaxy.
 2. Compare the theoretical velocity with observational data from galaxy datasets.
 3. Introduce the dark matter hypothesis as a potential solution to the discrepancy.
- Tools:** LIM, Cosmo-Edu-Lab App, spreadsheets, galaxy datasets.
- **Procedure:**
 1. Review of the law of gravitation (Newton), planetary motion in the solar system (Kepler), classical mechanics (uniform circular motion), and conservation laws (energy and angular momentum).
 2. Application of classical physics to stellar motion within galaxies.
 3. **Rotation Curve Exercise using App (Group/Pair work):**
 - The application loads a pre-simulated model of a spiral galaxy (e.g. :NGC3198). The interface visually displays the galaxy and a graph of orbital velocity versus distance with a red curve representing the Keplerian velocity, blue points that are real data and a green curve (simulation).

- The students observe that the red rotation curve decreases with distance (Keplerian behavior), in clear contrast to the real observational data displayed.
 - Using a slider to increase Dark Matter, initially set to zero, the student increases the density of the dark matter halo, and the green simulated curve updates in real-time until it "fits" the real data.
 - Sonification: By moving the slider, the student will hear a tone that drops in pitch (without dark matter) or remains high and constant (with dark matter), both for the curve and the mean, providing multimodal feedback (also for impaired students). Data are converted in sound frequency using a json library, high values are represented by high-pitched sound, low values by low-pitched sound.
4. **Chi-Squared Fitting:** Students insert three points of dark matter mass and the application will compute the interpolation of a parabola in order to find the minimum of the chi squared, which represents the best fit between observations and simulated curve.
 5. **Check evidence of DM:** Moving the slider to add dark matter, students can check also what is the correct quantity of dark matter necessary to fit the observation in the plot of mass vs radius. Moreover, a plot representing the structure of a galaxy from above shows that stars and gas (baryonic mass) is concentrated in the galaxy disk, while the dark matter is the composition of halo.
 6. **Debate:** "Why does the observed curve remain flat even at large distances from the galactic center?"; "If the velocity remains constant even at great distances, what does this imply for the total energy and angular momentum of the stars?" Goal: Stimulate critical reflection before the explicit introduction of dark matter.

• **Dataset variables of galaxies:** [15]

- Rad (radius)
- V_{obs} (observed velocity)
- $\text{err}V$ (velocity error)
- V_{gas} (gas velocity)
- V_{disk} (stellar disk velocity)
- V_{bul} (bulge velocity)
- SB_{disk} (surface brightness disk)
- SB_{bul} (surface brightness bulge)

Numerical and physical methods and models:

Classical Physics derivation:

- We model a small mass m (orbiting body) in a stable circular orbit around a large central mass M (e.g., galactic center) at a radius r .

- The condition for stable orbit requires the Gravitational Force (F_G) to equal the Centripetal Force (F_c).

1. Gravitational and Centripetal Forces

- **Gravitational Force (F_G):**

$$F_G = \frac{GMm}{r^2} \quad (1)$$

- **Centripetal Force (F_c):**

$$F_c = \frac{mv^2}{r} \quad (2)$$

- **Equating the Forces ($F_G = F_c$):**

$$\frac{GMm}{r^2} = \frac{mv^2}{r} \quad (3)$$

2. Deriving the rotational velocity (v)

- **Cancellation of m :** The mass of the orbiting body (m) cancels out, showing v is independent of m .

$$\frac{GM}{r^2} = \frac{v^2}{r} \quad (4)$$

- **Isolating v^2 :** Multiply by r to isolate v^2 .

$$v^2 = \frac{GM}{r} \quad (5)$$

- **Final Rotational Velocity (v):** Take the square root.

$$v = \sqrt{\frac{GM}{r}} \quad (6)$$

- **Physical Meaning:** This Keplerian velocity profile dictates that $v \propto 1/\sqrt{r}$, meaning orbital speed decreases as distance from the central mass increases.

3. Connection to Kepler's third law

- **Orbital Period (T):** Period is the circumference divided by velocity.

$$T = \frac{2\pi r}{v} \quad (7)$$

- **Kepler's relation :** Square T and substitute $v^2 = GM/r$.

$$T^2 = \frac{4\pi^2 r^2}{v^2} = \frac{4\pi^2 r^2}{GM/r} \implies T^2 = \frac{4\pi^2}{GM} r^3 \quad (8)$$

- **Physical Meaning:** This confirms Kepler's Third Law ($T^2 \propto r^3$), demonstrating that the proportionality derived from the force balance is fully consistent with the relationship between orbital period and radius.

Velocity and mass

- **Baryonic velocity (from data):**

$$V_{\text{baryons}} = \sqrt{V_{\text{gas}}^2 + V_{\text{disk}}^2 + V_{\text{bul}}^2} \quad (9)$$

- **Baryonic mass:**

$$M_{\text{baryons}}(r) = \frac{r V_{\text{baryons}}^2(r)}{G} \quad (10)$$

- **Total observed mass:**

$$M_{\text{obs}}(r) = \frac{r V_{\text{obs}}^2(r)}{G} \quad (11)$$

- **Dark matter velocity**, using the velocity components from data:

$$V_{\text{DM}} = \frac{V_{\text{obs}}^2(r)}{r} - \frac{V_{\text{gas}}^2(r)}{r} - \frac{V_{\text{disk}}^2(r)}{r} - \frac{V_{\text{bul}}^2(r)}{r} \quad (12)$$

- **Dark matter density (from data):**

$$\rho_{\text{DM}}(r) = \frac{1}{4\pi r^2} \frac{dM_{\text{DM}}}{dr} = \frac{1}{4\pi G r^2} \frac{d}{dr} (r^2 V_{\text{DM}}^2(r)) \quad (13)$$

NFW Dark Matter Profile [19, 18]

- **NFW profile for dark matter:**

$$\rho_{\text{NFW}}(r) = \frac{\rho_s}{(r/r_s)(1 + r/r_s)^2} \quad (14)$$

where r_s is the scale radius and ρ_s is the characteristic density, $\alpha = 1, \beta = 3, \gamma = 1$.

$$r_{200} = \left[\frac{3M_{200}}{4\pi 200\rho_c} \right]^{1/3} \quad (15)$$

with the critical density: $\rho_c = 2.775 \cdot 10^2 M_{\odot}/\text{kpc}^3$.

- The characteristic radius:

$$r_s = \frac{r_{200}}{C} \quad (16)$$

- The concentration parameter is:

$$C = 0.3 \cdot 11.7 \cdot \left[\frac{M_{200}}{10^{11} M_{\odot}} \right]^{-0.075} \quad (17)$$

- **Matching observed density and NFW:**

$$\rho_{\text{DM}}(r_{\text{match}}) = \rho_{\text{NFW}}(r_{\text{match}}) \quad (18)$$

From this, obtain M_{200} , r_{200} , and c . The concentration parameter $c = r_{200}/r_s$.

- **NFW dark matter mass:**

$$M_{\text{NFW}}(r) = 4\pi \int \rho_{\text{NFW}} r^2 dr = 4\pi \rho_s r_s^3 \left[\ln \left(1 + \frac{r}{r_s} \right) - \frac{r/r_s}{1 + r/r_s} \right] \quad (19)$$

- **Velocity dark matter:**

$$V_{\text{DM}}(r) = \sqrt{\frac{GM_{\text{NFW}}(r)}{r}} \quad (20)$$

- **Total simulated curve:**

$$V_{\text{sim}}(r, f) = \sqrt{\frac{GM_{\text{baryons}}(r) + f \cdot M_{\text{DM}}(r)}{r}} \quad \text{with } f \in [0, 2] \quad (21)$$

Data visualization and fit

- **Plot:**

- **X-axis:** Rad (radius, data)
- **Y-axis:**
 - * V_{baryons} (red)
 - * V_{obs} (data, blue) $\pm \text{errV}$
 - * V_{sim} (green)

- **Fit χ^2/dof (chi-squared):**

$$\chi^2/\text{dof} = \frac{1}{N - P} \sum_{i=1}^N \left(\frac{V_{\text{obs}}(r_i) - V_{\text{sim}}(r_i)}{\text{errV}(r_i)} \right)^2 \quad (22)$$

where N is the number of data points and P is the number of free parameters.

Detailed explanation from literature [12, 13, 15, 28, 16]

1) Centrifugal Equilibrium

$$\frac{V_{\text{obs}}^2(r)}{r} = a_H(r) + a_D(r) + a_{HI}(r) \quad (23)$$

where:

- $a_H(r)$ = radial acceleration due to DM halo
- $a_D(r)$ = radial acceleration due to stellar disk

- $a_{HI}(r)$ = radial acceleration due to gas disk

For a spherical halo (Eq. 8, [13]):

$$a_H(r) = \frac{4\pi G}{r^2} \int_0^r \rho_H(r') r'^2 dr' \quad (24)$$

2) Analytical derivation

Multiplying Eq. (23) by r^2 (Eq. 9 [13]):

$$r^2 V^2(r) = 4\pi G \int_0^r \rho_H(r') r'^2 dr' + r^2 (a_D(r) + a_{HI}(r)) \quad (25)$$

Taking the derivative with respect to r gives the DM halo density:

$$\rho_H(r) = \frac{1}{4\pi G r^2} \frac{d}{dr} \left[r^2 \left(\frac{V^2(r)}{r} - a_D(r) - \frac{V_{HI}^2(r)}{r} \right) \right] \quad (26)$$

3) Baryonic Accelerations

Stellar disk:

$$a_D(r) = \frac{V_{\text{disk}}^2(r)}{r} \quad (27)$$

Gas disk:

$$a_{HI}(r) = \frac{V_{\text{gas}}^2(r)}{r} \quad (28)$$

4) Numerical derivative

Define [13]:

$$q(r) = r^2 \left(\frac{V^2(r)}{r} - a_D(r) - \frac{V_{HI}^2(r)}{r} \right) \quad (29)$$

Then:

$$\rho_{DM}(r_i) \approx \frac{1}{4\pi G r_i^2} \left. \frac{dq}{dr} \right|_{r_i} \quad (30)$$

5) NFW profile

Density profile:

$$\rho_{\text{NFW}}(r) = \frac{\rho_s}{(r/r_s)(1 + r/r_s)^2} \quad (31)$$

Enclosed mass:

$$M_{\text{NFW}}(< r) = 4\pi \rho_s r_s^3 \left[\ln(1+x) - \frac{x}{1+x} \right], \quad x = \frac{r}{r_s} \quad (32)$$

Relations between parameters:

$$R_{200} = \left(\frac{3M_{200}}{4\pi \cdot 200 \cdot \rho_{\text{crit}}} \right)^{1/3} \quad (33)$$

$$r_s = \frac{R_{200}}{c} \quad (34)$$

$$\rho_s = \frac{M_{200}}{4\pi r_s^3 \left[\ln(1+c) - \frac{c}{1+c} \right]} \quad (35)$$

6) Observational matching

- Choose an external matching radius r_{match} (NGC 3198: $r_{\text{match}} \approx 44$ kpc).
- Compute $\rho_{\text{DM}}(r_{\text{match}})$ using Eq. (30).
- Determine M_{200} from the condition:

$$\rho_{\text{NFW}}(r_{\text{match}}; M_{200}) = \rho_{\text{DM}}(r_{\text{match}}) \quad (36)$$

Then recover:

$$c, \quad r_s, \quad \rho_s$$

from the parameter relations above.

1.2 Galaxy mass and velocity

- **Objectives:** Using Newtonian gravity, students will model the rotational velocity of stars in spiral galaxies to see how observed flat rotation curves contradict Keplerian dynamics. Students understand how Newtonian gravity predicts rotation speeds and how discrepancies lead to inference of invisible mass.
- **Dataset variables of galaxies:** [15]
- **Procedure:**
 1. Students begin by reviewing the classical derivation of orbital velocity $v(r) = \sqrt{GM(r)/r}$.
 2. **Galaxy Mass Calculation Exercise (Group/Pair work):** Students select the dataset of a galaxy (e.g. NGC 55) and complete the missing field of the pseudo-code to make the velocity and mass plots; they need to input the correct formulas of baryonic and total mass using symbols provided in the legend.
 3. If the formulas are correct the plots of baryonic VS total mass and velocities appear and the respective image of the galaxy.
 4. Compare the data with and without dark matter for different galaxies, verifying that the need of dark matter is common for all of them.
 5. **Guided calculation exercise:** In groups/pairs, calculate 2-3 Keplerian velocity values $v(r)$ starting from a dataset using the formula and compare them with the observed data values, using a spreadsheet. Goal: Evaluate the application of classical physics and identify discrepancy with observations.

Numerical and physical methods and models:

1. Compute the baryonic velocity as the quadrature sum of each components from data:

$$V_{\text{baryons}}(r) = \sqrt{V_{\text{gas}}^2(r) + V_{\text{disk}}^2(r) + V_{\text{bul}}^2(r)} \quad (37)$$

2. Derive the baryonic mass:

$$M_{\text{baryons}}(r) = \frac{rV_{\text{baryons}}^2(r)}{G} \quad (38)$$

3. Compute the total mass from observations:

$$M_{\text{obs}}(r) = \frac{rV_{\text{obs}}^2(r)}{G} \quad (39)$$

4. Compute the dark matter mass (assuming the NFW profile mass M_{NFW}):

$$M_{\text{DM}}(r) = 4\pi\rho_s r_s^3 \left[\ln \left(1 + \frac{r}{r_s} \right) - \frac{r/r_s}{1 + r/r_s} \right] \quad (40)$$

5. Plot X-axis Rad (radius, data), Y-axis:

- V_{baryons} (red)
- V_{obs} (data blue)
- V_{DM} (red)
- V_{sim} (blue)

1.3 Velocity distribution in galaxy clusters

- **Objectives:** Help students infer the existence of dark matter in galaxy clusters using classical mechanics and virial theorem.
- **Procedure:**
 1. Review circular motion of a light particle around a heavy one, law of gravitation, total energy, gravitational potential, and kinetic energy.
 2. Derive the Virial Theorem ($2K + U = 0$) from classical physics formulas (explanation with derivative of inertia momentum only for 5th grade if necessary). Derive the Virial mass in the galaxy cluster, introducing the concept of velocity dispersion.
 3. A pre-simulated model of a galaxy cluster (Coma) is uploaded and the application generates an histogram of cluster velocity distribution using a real dataset.
 4. The panel shows an histogram of galaxy velocities distribution from baryonic velocity (red), real data (blue) and a simulated histogram (green).
 5. The mass tuning slider allows students to increase the dark matter mass in the cluster, visualizing the change in the galaxy velocities of the green histogram in real-time until it overlaps the observations.

6. Other visualizations: View a scatter plot of galaxy velocity vs. radius and an animation of galaxy movement (comparison of observations and simulation).
7. Sonification: By moving the slider, the student will hear a tone that drops in pitch (without dark matter) or remains high and constant (with dark matter) for the mean of galaxy velocity distribution.

Numerical and physical methods and models: Dataset Coma cluster (SDSS) : objid (galaxy ID), ra (right ascension, α), dec (declination, δ), modelmag (apparent magnitude in r band, m_r), modelmagerr, extinction, redshift (z), zErr (redshift error) [31, 2].

1. Velocity of each galaxy from redshift (data):

$$v_i = c z_i \quad (41)$$

2. Number of observed galaxies: N

3. Velocity mean:

$$\bar{v} = \frac{1}{N} \sum_{i=1}^N v_i \quad (42)$$

4. Observed velocity dispersion:

$$\sigma_{\text{obs}} = \sqrt{\frac{1}{N} \sum_{i=1}^N (v_i - \bar{v})^2} \quad (43)$$

5. Comoving distance:

$$\chi(z) = \frac{c}{H_0} \int_0^z \frac{dz'}{\sqrt{\Omega_m(1+z')^3 + (1-\Omega_m)}} \quad (44)$$

6. Angular diameter distance:

$$D_A(z) = \frac{\chi(z)}{1+z} \quad (45)$$

7. Search the cluster centers from BCG or median coordinates:

$$(\alpha_c, \delta_c) = (\alpha_{\text{BCG}}, \delta_{\text{BCG}}) \quad \text{or} \quad (\text{median}(\alpha), \text{median}(\delta))$$

8. Angular separation:

$$\theta = \arccos(\sin \delta_1 \sin \delta_2 + \cos \delta_1 \cos \delta_2 \cos(\alpha_1 - \alpha_2)) \quad (46)$$

9. Projected radius:

$$r_{\text{proj},i} = \theta_i D_{A,\text{kpc}} \quad (47)$$

10. **Critical density:**

$$\rho_{\text{crit}} = \frac{3H_0^2}{8\pi G} \quad (48)$$

11. **Virial relations:**

$$\begin{aligned} \sigma_{\text{obs}}^2 &= \frac{GM_{200}}{3R_{200}} \\ M_{200} &= \frac{4}{3}\pi 200 \rho_{\text{crit}} R_{200}^3 \end{aligned} \quad (49)$$

Dark Matter (NFW) Profile

12. **Concentration (fit relation):**

$$c = A \left(\frac{M_{200}}{M_{\text{pivot}}} \right)^B (1+z)^C \quad (50)$$

13. **NFW density factor:**

$$\delta_c(c) = \frac{200}{3} \frac{c^3}{\ln(1+c) - c/(1+c)} \quad (51)$$

14. **Characteristic radius:**

$$r_{200} = \left(\frac{3M_{200}}{4\pi 200 \rho_{\text{crit}}} \right)^{1/3} \quad (52)$$

15. **Characteristic density and scale radius:**

$$\rho_s = \delta_c(c) \rho_{\text{crit}}, \quad r_s = \frac{r_{200}}{c}$$

16. **Dark matter mass (NFW):**

$$M_{\text{NFW}}(r_{\text{proj},i}) = M_{200} \frac{\ln(1+x_i) - x_i/(1+x_i)}{\ln(1+c) - c/(1+c)}, \quad x_i = \frac{r_{\text{proj},i}}{r_s} \quad (53)$$

Luminous Mass

17. **Luminosity distance (small z):**

$$D_L \approx \frac{cz_{\text{cluster}}}{H_0}, \quad z_{\text{cluster}} = \text{median}(z_i) \quad (54)$$

18. **Distance modulus:**

$$\text{distmod} = 5 \log_{10}(D_L) - 5 \quad (55)$$

19. **Absolute magnitude:**

$$M_r = m_r - A_r - \left[5 \log_{10} \left(\frac{D_L}{10 \text{ pc}} \right) \right] \quad (56)$$

20. **Luminosity:**

$$L_r = 10^{0.4(M_{r\odot} - M_r)} \quad (57)$$

21. **Stellar/baryonic mass per galaxy:**

$$M_{\text{bar},i} = (M/L) L_r, \quad M/L = 2 \quad (58)$$

22. **Total baryonic mass:**

$$M_{\text{bar}} = \sum_{i=1}^N M_{\text{bar},i} \quad (59)$$

23. **Characteristic baryonic radius:**

$$r_{\text{bar}} = \left(\frac{3M_{\text{bar}}}{4\pi 200 \rho_{\text{crit}}} \right)^{1/3} \quad (60)$$

24. **Velocity dispersion (baryons only):**

$$\sigma_{\text{bar}} = \sqrt{\frac{GM_{\text{bar}}}{3r_{\text{bar}}}} \quad (61)$$

Velocity Dispersion and Simulation

25. **Total simulated mass:**

$$M_{\text{tot}} = M_{\text{bar}} + f M_{200} \quad (62)$$

26. **Total simulated velocity dispersion:**

$$\sigma_{\text{sim}} = \sqrt{\frac{GM_{\text{tot}}}{3r_{200}}} \quad (63)$$

27. **Observed histogram:**

$$H_{\text{obs}} = \text{hist}(v_{\text{obs}})$$

28. **Generate normal distribution of simulated velocities:**

$$\sigma_{\text{tot}} = \text{Normal}(\mu = \sigma_{\text{sim}}, \sigma = \sigma_{\text{obs}}, N)$$

29. **Simulated histogram:**

$$H_{\text{sim}} = \text{hist}(\sigma_{\text{tot}})$$

Data visualization

- Plot histograms: number of galaxies on Y-axis, velocity on X-axis.
- Curves: observations (blue), simulated (green), baryonic (red).

1.4 Cluster Mass via Virial Theorem

- **Objectives:** Students will apply the virial theorem to estimate the total mass of a galaxy cluster from the kinetic and potential energy of its galaxies.

- **Procedure:**

1. Introducing the virial theorem $2K + U = 0$.
2. **Calculation Exercise (Group/Pair work):** Students select a dataset of a cluster of galaxies and complete the missing field of the pseudo-code to make the mass and density plots; they need to input the correct formulas of virial/total mass using symbols provided in the legend.
3. If the formulas are correct the plots of luminous VS total mass and density appear.
4. Compare the data with and without dark matter and find the necessity of extra mass in different clusters.
5. **Challenge Quiz:** Multiple-choice questionnaire (6–8 mixed questions) in challenge mode for groups: application of formulas, key concepts. Goal: Monitor the understanding of concepts, data, and physical quantities.
6. **Debate:** Guided group discussion on the meaning of the results, comparison between galaxy and cluster models.

Numerical and physical methods and models:

Dataset Abell Clusters: Cluster (name), ID (galaxy ID), RAdeg (right ascension in degrees), DEdeg (declination in degrees), RV (radial velocity in km/s), e_RV (velocity error in km/s), q_RV (quality flag), bmag (apparent magnitude in B band). [35]

Cluster kinematics

1. **Angular Separation ($\Delta\theta$)** from right ascension (α) and declination (δ) data (converted to radians, where c denotes the cluster center and i the individual galaxy):

$$\Delta\theta = \arccos[\sin(\delta_c)\sin(\delta_i) + \cos(\delta_c)\cos(\delta_i)\cos(\alpha_c - \alpha_i)] \quad (64)$$

2. **Projected Radius (R_P)** from angular separation, assuming a distance $D = 100$ Mpc (radius converted to kpc):

$$R_P = D \cdot \Delta\theta \quad (65)$$

3. **Observed Velocity (V_{obs})** from redshift data (or radial velocity RV):

$$V_{\text{obs}} = c \cdot z \quad \text{or} \quad V_{\text{obs}} = \text{RV} \quad (66)$$

4. **1-D Velocity Dispersion (σ_{1D})** (observation):

$$\sigma_{1D} = \sqrt{\frac{1}{N} \sum_{i=1}^N (V_i - \bar{V})^2} \quad (67)$$

5. **3-D Velocity Dispersion (σ_{3D})** (assuming isotropy):

$$\sigma_{3D} = 3 \sigma_{1D} \quad (68)$$

Baryonic components

6. **Convert Distance D (in Mpc) to pc** and compute the **Distance Modulus (μ)**:

$$\mu = 5 \log_{10} \left(\frac{D_{\text{pc}}}{10 \text{ pc}} \right) \quad (69)$$

7. **Absolute Magnitude (M)** using apparent magnitude ($m = \text{bmag}$):

$$M = m - \mu \quad (70)$$

8. **Luminosity (L)** of each galaxy from absolute magnitude (M):

$$L = 10^{-0.4(M-M_{\odot})} \quad (71)$$

9. **Total Luminosity of the cluster (L_{total})**:

$$L_{\text{total}} = \sum_{i=1}^N L_i \quad (72)$$

10. **Luminous/Baryonic Mass (M_{lum})** from luminosity (Υ is the mass-to-light ratio):

$$M_{\text{lum}} = \Upsilon \cdot L_{\text{total}} \quad (73)$$

Virial Theorem Derivation

11. **Gravitational Potential Energy** (light particle m orbiting heavy particle M):

$$U = -\frac{GMm}{r} \quad (74)$$

12. **Gravitational Force**:

$$F_G = \frac{GMm}{r^2} \quad (75)$$

13. **Centripetal Force**:

$$F_c = \frac{mv^2}{r} \quad (76)$$

14. **Equating Forces** ($F_G = F_c$) to find v^2 :

$$v^2 = \frac{GM}{r} \quad (77)$$

15. **Kinetic Energy** (K , neglecting heavy particle's kinetic energy):

$$K = \frac{1}{2}mv^2 \quad (78)$$

16. **Replacing v^2 in K :**

$$K = \frac{1}{2}m \left(\frac{GM}{r} \right) \quad (79)$$

17. **Replacing Potential Energy U in K :**

$$K = -\frac{1}{2}U \quad (80)$$

18. **Virial Theorem:**

$$2 \langle K \rangle + \langle U \rangle = 0 \quad (81)$$

Cluster Virial Mass and Density

- **Context:** In a cluster, the total potential energy is the sum of all pairwise interactions, and the total kinetic energy is the sum of the kinetic energies of all galaxies. The Virial Theorem relates these totals. For an isotropic and stable system, the velocity is replaced by the velocity dispersion (σ).

19. **Total Virial Mass of the cluster** (M_{Virial}) (using σ_{3D} and characteristic radius R):

$$M_{\text{Virial}} = \frac{3\sigma_{1D}^2 R}{G} \quad (82)$$

20. **Total Volume** (approximated as a sphere):

$$\text{Volume} = \frac{4}{3}\pi R^3 \quad (83)$$

21. **Luminous Density** (ρ_{lum}):

$$\rho_{\text{lum}} = \frac{M_{\text{lum}}}{\text{Volume}} \quad (84)$$

22. **Total Density** (ρ_{total}):

$$\rho_{\text{total}} = \frac{M_{\text{total}}}{\text{Volume}} \quad (85)$$

Data visualization

- **X-Axis:** Radius (R).

- **Y-Axis (Quantities to Plot):**
 1. ρ_{lum} (Luminous Density, Red)
 2. ρ_{total} (Total Density, Data Blue)
 3. V_{sim} (Total Simulated Velocity, Red)
 4. V_{obs} (Observed Velocity, Blue data points)

Phase 3: Elaborate - Authentic Task

- **Tools:** Calculator, spreadsheet, datasets, PC.
- **Authentic Task:** A multimedia dossier "In Search of Dark Matter" (to be presented in PowerPoint, poster, or digital report format) that includes:
 - **Galaxy Rotation Curves:** Given a dataset with the average radius (r in kpc), the observed velocity, and the luminous mass contained within r (in $M_{\odot}$), use a spreadsheet to perform the following exercises:
 - * Calculate the expected orbital velocity $V(r)$ using Newton's law for at least 5 values of r .
 - * Plot the Keplerian $V(r)$ graph.
 - * Plot the observed curve on the same graph, compare the two curves and describe the discrepancy, quantitatively estimate how much dark mass would be needed to explain the observed curve.
 - **Galaxy Cluster and Virial Theorem:** A dataset is provided with the velocities of 10 galaxies in a cluster (in km/s), the average radius of the cluster (in Mpc), and the luminous mass.
 - * Calculate the velocity dispersion of the cluster.
 - * Apply the Virial Theorem to estimate the total mass of the cluster.
 - * Compare with the given luminous mass and calculate the total mass/luminous mass ratio.
 - * Argue in 5–6 lines what the result implies.
 - **Conceptual Schema/Map:** Synthesis of the journey: from classical laws $\rightarrow$ observations $\rightarrow$ evidence of dark matter. Find an analogy (e.g., iceberg).
 - **Open Questions:**
 - * "Why is classical mechanics insufficient to explain cosmological observations?"
 - * "Newton's law of universal gravitation perfectly described the motion of planets, but in cosmology it leads to discrepancies with data. What does this teach us about the scientific method and the evolution of theoretical models?"

- **Inclusion and Differentiation:** Students with PDP (Personalized Didactic Plan) or PEI (Individualized Educational Plan) will benefit from personalized activities with adapted materials, reduced datasets, pre-set formulas, and flexible timing, consistent with the principles of inclusive teaching and Universal Design for Learning (UDL). To ensure the participation of all students, the authentic task will be offered at two levels of difficulty:
 - **Basic Level:** Simplified data analysis, guided formulas, and visual supports, intended for students with BES (Special Educational Needs) or calculation difficulties.
 - **Advanced Level:** Complete analysis, autonomous data processing, and argumentative depth in the dossier.

Phase 4: Assessment

- Presentations of group work, collective reflection.
- Peer assessment (groups) in oral form.
- Self-assessment: metacognitive reflection sheet to develop learning awareness.

2 Universe History and CMB

Thermodynamics → Universe Expansion/Evolution and CMB

- **Classical physics:** First law of thermodynamics, adiabatic transformations, thermal behavior of gases under expansion (monoatomic 5/3, diatomic 7/5), entropy, Planck's law, blackbody spectrum, photons, Stefan-Boltzmann law, Doppler effect, reversible and irreversible processes, Einstein's energy.
- **Cosmology:** expansion and evolution (thermal history) of the universe, matter and radiation, CMB, redshift, scale factor.

Classical Physics and Thermodynamics [4]

- **First Law of Thermodynamics:** The change in internal energy dU of a system equals the heat added to the system dQ minus the work done by the system dW .

$$dU = dQ - dW \quad (86)$$

- **Adiabatic Transformation:** A process where no heat is exchanged ($dQ = 0$). In this case, the total internal energy changes only due to work performed.

$$dU = -dW \quad \text{and} \quad PV^\gamma = \text{constant} \quad (87)$$

where P is pressure, V is volume, and γ is the adiabatic index (or ratio of specific heats).

- **Adiabatic Index (γ):** Describes the thermal behavior of a fluid under expansion.
 - Monoatomic gas: $\gamma = 5/3$
 - Diatomic gas: $\gamma = 7/5$
 - Radiation (photons, applicable to the early Universe): $\gamma = 4/3$
- **Blackbody Spectrum (Planck's Law):** The spectral radiance $B_\nu(T)$ describes the intensity of radiation emitted by a black body at temperature T and frequency ν .

$$B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/k_B T} - 1} \quad (88)$$

- **Stefan-Boltzmann Law:** The total energy density of blackbody radiation ρ_{rad} is proportional to the fourth power of the temperature.

$$\rho_{\text{rad}} = \alpha T^4 \quad \text{where} \quad \alpha = \frac{4\sigma}{c} \quad (89)$$

- **Wien's Displacement Law:** The peak wavelength λ_{peak} of blackbody radiation is inversely proportional to the temperature T . This law is used for a counter-check to find the temperature from the peak wavelength.

$$\lambda_{\text{peak}} \cdot T = b \quad (90)$$

where b is Wien's displacement constant.

Cosmology context:

- **Cosmological Expansion:** The expansion and thermal history of the Universe lead to distinct epochs: radiation-dominated, matter-dominated, and dark energy-dominated.
- **Cosmic Microwave Background (CMB):** Isotropic radiation discovered by Penzias and Wilson, characterized by an almost perfect blackbody spectrum, with small fluctuations representing density variations.
- **Redshift and Scale Factor ($a(t)$):** The observed cosmological redshift z is related to the expansion factor of the Universe, $a(t)$, where t_0 is the present time:

$$1 + z = \frac{a(t_0)}{a(t)} \quad (91)$$

- **Present CMB Temperature (T_0):** The CMB currently has a temperature of $T_0 \approx 2.725$ K.

2.1 CMB blackbody spectrum and T0

[9, 30, 1]

Objectives:

- Review the Planck's law for a black body and connect to the CMB spectrum,
- Reproduce the curve wavelength vs radiation intensity for different values of temperature peaks,
- Find the value of the temperature of CMB at present time comparing the reproduced theoretical curves with the CMB spectrum until the correct curve overlaps the data.

Dataset: COBE/FIRAS CMB monopole: wavelengths, radiation intensity [6, 9, 8] **Procedure:**

- The CMB can be considered as a blackbody spectrum, with photons in thermal equilibrium with the plasma and small fluctuations representing density variations. As the universe expands, the temperature decreases and the photon wavelength increases until it reaches the microwave band in the spectrum.
- Find the present-day temperature T_0 , starting from Planck's law for the blackbody spectrum:

$$B_\lambda(T) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{\frac{hc}{\lambda kT}} - 1} \quad (92)$$

- Choose several temperature values (peaks) and define a wavelength range. Use one of the following online calculators to derive the corresponding radiation intensities: [34, 21, 3, 33]
- Create a table with radiation intensity versus wavelength/frequency for each selected peak temperature.
- Plot radiation intensity (y-axis) versus wavelength (x-axis) for each table, each characterized by a different peak temperature (e.g. 3000 K, 5000 K, 10000 K).
- Compare the theoretical blackbody spectrum curves with the observed CMB blackbody spectrum at the present-day temperature $T_0 = 2.725$ K from the COBE/FIRAS CMB monopole dataset: wavelengths, radiation intensity [6, 9, 8].
- Measure the value of the spectral peak $\lambda_{\max}$ from the theoretical curves (at different temperatures) and compare it with the COBE CMB spectrum peak $\lambda_{\max,0} \approx 1.06$ mm.
- Verify that the peaks of the theoretical spectra are shifted with respect to the CMB spectrum. If the temperature is higher, the curve has a higher and left-shifted peak (toward shorter wavelengths). The distance between peaks indicates the cooling (toward longer wavelengths).
- As a consistency check, recall Wien's displacement law, which analytically gives the wavelength of the spectral peak:

$$\lambda_{\max} = \frac{b}{T} \quad (93)$$

with $b = 2.89 \times 10^{-3}$ m K. By finding $\lambda_{\max,0}$ from the plot, one can infer the corresponding temperature.

- Determine the curve that best represents the CMB Planck spectrum, and thus the correct value of the present-day temperature T_0 .

2.2 Thermodynamics and adiabatic expansion

[17, 27, 36, 7, 14]

Objectives:

- Review first thermodynamic law, adiabatic expansion and the energy equipartition theorem for monoatomic and biatomic gas,
- Find the correct value of the adiabatic index γ for photons (radiation) that characterized the primordial universe,
- Move the slider γ to find the best match of the simulated curve with data in the plot temperature vs redshift,
- Find that the radiation is composed of particles without mass (photons) and the universe is not static (expansion).

Dataset: CMB temperatures, redshift. [24] **Procedure:**

- Imagine the universe on large scales as isotropic and homogeneous, a closed and isolated system (no heat or energy exchange with the “outside”). We can approximate that the total internal energy is conserved over time; according to Einstein’s relativity, energy and mass are therefore conserved independently.
- The primordial universe was hot, opaque, and dominated by photon radiation.
- Penzias and Wilson discovered that the universe contains a cosmic microwave background (CMB), an isotropic signal in the microwave band with the same intensity in all directions, whose local variations differ from the mean by only about 10^{-5} .
- The energy spectrum of the CMB is a perfect blackbody (with $T_0 = 2.726$ K at the present epoch), and its wavelength is inversely proportional to temperature.
- The temperature of the universe decreased over time inversely with its size, proportional to cosmic expansion.
- Hence, the universe’s expansion can be regarded as an adiabatic process: an isolated system, conserved total energy, expansion with time, and temperature decrease as volume increases.

First law of thermodynamics:

$$\Delta U = Q - L \quad \Rightarrow \quad dQ = dU + P dV \quad (94)$$

For an adiabatic process $dQ = 0$:

$$dU + P dV = 0 \quad (95)$$

Adiabatic transformation:

$$Q = 0 \quad \Rightarrow \quad L = -\Delta U = C_v n \Delta T \quad (96)$$

$$PV = nkT \quad \Rightarrow \quad PV^\gamma = \text{const} \quad (97)$$

$$TV^{\gamma-1} = \text{const} \quad \Rightarrow \quad T = V^{1-\gamma} \quad (98)$$

$$\gamma = \frac{C_p}{C_v} \quad (99)$$

Energy equipartition theorem:

$$\langle E \rangle = \frac{\int (ax^2) e^{-\beta ax^2} dx}{\int e^{-\beta ax^2} dx} = \frac{k_B T}{2} \quad (100)$$

where the probability $P(x)$ of the system to have energy $E = ax^2$ is proportional to the Boltzmann distribution factor $e^{-\beta ax^2}$, with $\beta = 1/(k_B T)$. The denominator is a normalization with probability equal to one. The energy is the sum of n quadratic degree of freedom called modes of the system.

$$\langle E \rangle = n \cdot 1/2 k_B T \quad (101)$$

The energy is distributed to all the system modes. **Energy equipartition applications:**

- Monoatomic gas (3 dof), involves only translations:

$$K = \frac{1}{2} m (v_x^2 + v_y^2 + v_z^2) = 3 \cdot \frac{1}{2} k_B T = \frac{3}{2} k_B T, \quad U = \frac{3}{2} n R T, \quad C_v = \frac{3}{2} R, \quad C_p = \frac{5}{2} R, \quad \gamma = \frac{5}{3} \quad (102)$$

- Diatomic gas (5 dof), translations and rotations:

$$K = \frac{1}{2} m (v_x^2 + v_y^2 + v_z^2) + \frac{L_x^2}{2I_x} + \frac{L_y^2}{2I_y} = \frac{5}{2} k_B T, \quad U = \frac{5}{2} n R T, \quad C_v = \frac{5}{2} R, \quad C_p = \frac{7}{2} R, \quad \gamma = \frac{7}{5} \quad (103)$$

- Polyatomic gas, translation, rotations and vibrations:

$$K = \frac{1}{2} m (v_x^2 + v_y^2 + v_z^2) + \frac{L_x^2}{2I_x} + \frac{L_y^2}{2I_y} + \frac{1}{2} m V_r^2 + \frac{1}{2} k r^2 = \frac{7}{2} k_B T \quad (104)$$

- Polyatomic gas (6dof) with 3 rotational axis:

$$K = \frac{1}{2} m (v_x^2 + v_y^2 + v_z^2) + \frac{1}{2} (I_x \omega_x^2 + I_y \omega_y^2 + I_z \omega_z^2) = 6 \left(\frac{1}{2} k_B T \right) = 3 k_B T, \quad U = 3 n R T, \quad C_v = 3 R, \quad C_p = 4 R, \quad \gamma = \frac{4}{3} \quad (105)$$

In cosmology:

$$dU + P dV = 0 \quad \Rightarrow \quad \frac{dU}{dt} = -P(t) \frac{dV}{dt} \quad (106)$$

Stefan–Boltzmann law for radiation energy density:

$$u = \frac{U}{V} = \alpha_{\text{rad}} T^4 \quad (107)$$

where

$$\alpha_{\text{rad}} = \frac{4\sigma}{c} = 7.56 \times 10^{-16} \text{ J m}^{-3} \text{ K}^{-4} \quad (108)$$

$$U = \alpha_{\text{rad}} T^4 V, \quad P = \frac{1}{3} \alpha_{\text{rad}} T^4 \quad (109)$$

Thus:

$$\frac{1}{T} \frac{dT}{dt} = -\frac{1}{3V} \frac{dV}{dt} \quad (110)$$

Scale factor and photon wavelength:

$$a(t) \quad \Rightarrow \quad \lambda \propto a(t) \quad (111)$$

The photon wavelength increases with the scale factor (expansion $\rightarrow$ longer wavelengths $\rightarrow$ lower photon energy):

$$E = \frac{hc}{\lambda} \quad (112)$$

Expanding universe:

$$V \propto a^3 \quad \Rightarrow \quad \frac{d}{dt}(\ln T) = -\frac{d}{dt}(\ln a) \quad (113)$$

$$T(t) \propto a(t)^{-1} \quad (114)$$

Doppler effect $\rightarrow$ Redshift:

$$1 + z = \frac{\lambda_{\text{obs}}}{\lambda_{\text{em}}} \quad (115)$$

$$a = \frac{1}{1 + z} \quad (116)$$

$$T(z) = T_0(1 + z) \quad (117)$$

For an adiabatic relation $T = V^{1-\gamma}$:

$$T \propto a^{3(1-\gamma)} \Rightarrow T \propto (1+z)^{3(\gamma-1)} \quad (118)$$

For radiation:

$$\gamma = \frac{4}{3} \Rightarrow T \propto (1+z)^{3(\frac{4}{3}-1)} = (1+z) = \frac{1}{a} \quad (119)$$

- **Dataset CMB:** CMB temperatures, redshift. [24]
- Plot $T(z) = T_0(1+z)$, CMB temperatures at different redshifts from the dataset; $T_0 = 2.725$ K is the present CMB temperature.
- Include a slider for γ (adiabatic index) related to the law $T \propto (1+z)^{3(\gamma-1)}$.
- Which value of γ matches the observed CMB data? $\gamma = \frac{4}{3}$.
- Complete the correct formulas in the box in order to review the connections between classical physics (thermodynamic laws, adiabatic transformations) and cosmology (the universe radiation, adiabatic expansion).

2.3 Radiation and Matter equivalence

[7, 17, 36, 14, 27] **Objectives:**

- Review energy, Einstein's law for photons and Stefan-Boltzmann law,
- Derive the equations for matter and radiation in function of the redshift,
- Plot the curves radiation/matter energy density vs redshift/time,
- Find the correct value of the redshift and temperature for the equivalence epoch (radiation-matter),
- Discover that in the past the universe was characterized first by a radiation-dominated phase, than by a matter-dominated phase.

Procedure:

- After the Big Bang, the universe experienced an era dominated by radiation, followed by an epoch of "equivalence," where the densities of radiation and matter were equal, and then by a matter-dominated era.
- Radiation energy density decreases more rapidly than matter because photons lose energy due to the stretching of wavelengths during the universe's expansion.
- Consider a portion of the universe as a sphere of radius R , with the expansion defined by the scale factor $a(t)$, such that:

$$V \propto R^3 \propto a^3 \quad (120)$$

The energy density is:

$$u = \frac{U}{V} \quad (121)$$

- Complete the correct formulas in order to derive the radiation and matter density curves.
- Plot the matter and radiation densities (different units) as a function of redshift z , time (Myr) or expansion factor $a(t)$.

Matter:

- The energy of non-relativistic matter particles is:

$$E = mc^2 + \frac{1}{2}mv^2 \quad (122)$$

Neglecting the kinetic term, the total energy grows with the number of particles contained in the considered volume:

$$\rho_m = \frac{E}{V} \propto \frac{1}{a^3} \quad (123)$$

$$\rho_m = \Omega_m \rho_c \quad \Rightarrow \quad \rho_m(a) = \rho_{m,0} a^{-3} = \rho_{m,0} (1+z)^3 \quad (124)$$

- The present-day matter density is:

$$\rho_{m,0} = \Omega_{m,0} \rho_c \quad (125)$$

where the critical density is:

$$\rho_c = \frac{3H_0^2}{8\pi G} \approx 9.2 \times 10^{-27} \text{ kg m}^{-3} \quad (126)$$

with

$$H_0 = 70 \frac{\text{km/s}}{\text{Mpc}} = 2.27 \times 10^{-18} \text{ s}^{-1}, \quad \Omega_{m,0} = 0.315 \quad (127)$$

Radiation:

- For photons:

$$E_\gamma = h\nu = \frac{hc}{\lambda} \quad (128)$$

During expansion:

$$\lambda \propto a \quad \Rightarrow \quad E_\gamma \propto \frac{1}{a} \quad (129)$$

and the number density of photons decreases as:

$$n_\gamma \propto \frac{1}{a^3} \quad (130)$$

Each photon loses energy as its wavelength stretches (cosmological redshift). Thus:

$$\rho_r = n_\gamma E_\gamma \propto a^{-3} a^{-1} = a^{-4} \quad (131)$$

- Using the Stefan–Boltzmann law:

$$\rho_r = \alpha_{\text{rad}} T^4 \quad (132)$$

and from adiabatic expansion $T \propto 1/a$, we obtain:

$$\rho_r \propto T^4 \propto a^{-4} \quad (133)$$

$$\rho_r = \Omega_r \rho_c \quad \Rightarrow \quad \rho_r(a) = \rho_{r,0} a^{-4} = \rho_{r,0} (1+z)^4 \quad (134)$$

- From the Stefan–Boltzmann law:

$$\rho_{r,0} = \alpha_{\text{rad}} T_0^4 \quad \Rightarrow \quad \Omega_{r,0} = \frac{\rho_{r,0}}{\rho_c} \quad (135)$$

Matter–Radiation Equality:

- From the plot, find the redshift $z_{\text{eq}} = 3400$ where matter and radiation energy densities are equal.
- Verify this with the analytical condition:

$$\rho_m(z_{\text{eq}}) = \rho_r(z_{\text{eq}}) \quad (136)$$

$$\rho_{m,0} (1+z_{\text{eq}})^3 = \rho_{r,0} (1+z_{\text{eq}})^4 \quad (137)$$

$$1+z_{\text{eq}} = \frac{\rho_{m,0}}{\rho_{r,0}} = \frac{\Omega_{m,0}}{\Omega_{r,0}} \quad (138)$$

- Determine the temperature at which matter–radiation equality occurred:

$$T(z_{\text{eq}}) = T_0 (1+z_{\text{eq}}) \quad (139)$$

$$T_{\text{eq}} = 9267 \text{ K} \quad (140)$$

3 Redshift and universe expansion

Optics, Waves, Relativity $\rightarrow$ Redshift, Hubble, Expansion of the Universe

- **Classical physics:** waves, optics, Doppler effect, spectroscopy, redshift, wavelength, flux, luminosity, spectral lines, relativistic velocity, atomic transitions.
- **Cosmology:** cosmological redshift, Hubble law, galaxy recession, luminosity distance, magnitude, expansion of the universe, Λ CDM model.

Optics and Light Waves: [4]

- Radiant flux per unit area (radiance):

$$F = \frac{d\Phi}{dA}, \quad \Phi = \frac{dE}{dt} \quad (141)$$

where Φ is the total emitted power (radiant flux, i.e. energy per unit time).

- Assuming isotropic emission with spherical geometry:

$$A = 4\pi r^2 \quad (142)$$

The flux becomes:

$$F = \frac{\Phi}{4\pi r^2} \quad (143)$$

- The flux expressed as a function of wavelength:

$$F(\lambda) = \frac{dE}{dA dt d\lambda_{obs}} \quad (144)$$

- In cosmology, Φ is written as intrinsic luminosity L :

$$F = \frac{L}{4\pi d_L^2} \quad (145)$$

- In terms of wavelength:

$$F(\lambda) = \frac{dL(\lambda)}{4\pi d_L^2 d\lambda} \quad (146)$$

- Observed flux considering redshift:

$$F(\lambda_{obs}) = \frac{1}{1+z} \frac{L(\lambda_{emit})}{4\pi d_L^2} \quad (147)$$

Doppler Effect:

- For sound waves: v is the speed of sound in air (340 m/s).
- Observed frequency:

$$f_{obs} = f_{emit} \frac{v \pm v_r}{v \mp v_s} \quad (148)$$

where v is the wave velocity in the medium, v_r is the receiver velocity, and v_s is the source velocity.

- If receiver and source approach:

$$\frac{v + v_r}{v - v_s} \quad (149)$$

- If receiver approaches and source recedes:

$$\frac{v + v_r}{v + v_s} \quad (150)$$

- If both receiver and source recede:

$$\frac{v - v_r}{v + v_s} \quad (151)$$

- If receiver recedes and source approaches:

$$\frac{v - v_r}{v - v_s} \quad (152)$$

- If the source approaches a stationary observer:

$$f_{obs} = f_{emit} \frac{v}{v - v_s} \quad (153)$$

- If the source recedes from a stationary observer:

$$f_{obs} = f_{emit} \frac{v}{v + v_s} \quad (154)$$

- If the observer approaches a stationary source:

$$f_{obs} = f_{emit} \frac{v + v_r}{v} \quad (155)$$

- If the observer recedes from a stationary source:

$$f_{obs} = f_{emit} \frac{v - v_r}{v} \quad (156)$$

Doppler effect for light:

- For light waves, $v = c$, the speed of light, which is independent of the medium (unlike sound).

- If the source approaches the observer (stationary):

$$f_{obs} = f_{emit} \frac{v}{v - v_s}, \quad f_{obs} > f_{emit} \quad (157)$$

- If the source recedes from the observer:

$$f_{obs} = f_{emit} \frac{v}{v + v_s}, \quad f_{obs} < f_{emit} \quad (158)$$

- In the relativistic case, the speed of light is the same in every inertial reference frame, independent of whether the source is moving or stationary.
- If the observer's reference frame recedes from the source:

$$f_{obs} = f_{emit} \sqrt{\frac{1 - v/c}{1 + v/c}} \quad (159)$$

- If the observer's reference frame approaches the source:

$$f_{obs} = f_{emit} \sqrt{\frac{1 + v/c}{1 - v/c}} \quad (160)$$

Redshift:

- Definition of redshift:

$$z = \frac{f_{obs}}{f_{emit}} - 1 = \frac{\lambda_{obs} - \lambda_{emit}}{\lambda_{emit}} \quad (161)$$

thus:

$$\lambda_{obs} = (1 + z)\lambda_{emit} \quad (162)$$

- **Redshift:** when a source moves away from the observer, the received light has a longer wavelength than at emission, this corresponds to a shift toward the red in the visible spectrum (lower frequency end).
- **Blueshift:** when a source approaches the observer, the received light has a shorter wavelength than at emission, this corresponds to a shift toward the blue in the visible spectrum (higher frequency end).

3.1 Cosmological redshift

[5, 27, 23] **Objectives:**

- Review Doppler effect, redshift, flux, Einstein energy law, relativity,
- Identify the measured peak wavelength λ_{obs} in a selected galaxy spectrum wavelength vs flux,
- Compute the redshift using the standard λ_0 values for some selected emission lines,
- Discover that galaxies are redshifted (their spectra are shifted toward longer wavelengths),
- Compute and compare the non-relativistic and relativistic velocities vs redshift.

Dataset: SDSS galaxy spectra [32]: flux, spectral lines with measured peak observed wavelength λ_{obs} , redshift.

Atomic transitions (spectral lines) catalog: NIST database — atomic spectroscopy λ_0 . [20]
Procedure:

- Atoms have quantized energy levels. When an electron transitions between two levels, it emits or absorbs a photon with a fixed wavelength:

$$E = \frac{hc}{\lambda_0} \quad (163)$$

where λ_0 is the laboratory wavelength, the same everywhere in the Universe. Thus, spectral lines act as “cosmic fingerprints”: if you know the atom, you know its exact rest-frame wavelength.

- Elements such as hydrogen and helium are abundant in the universe, while other heavier elements (metals) are found in stars and galaxies. Galaxies contain abundant hydrogen, helium, oxygen, and metals. Therefore their spectra always show well-known lines such as $H\alpha$, $H\beta$, [O III], Ca II, etc.
- Common laboratory spectral lines: [20]
 - $H\alpha$ (6562.8 Å): Hydrogen transition $n = 3 \rightarrow n = 2$ (Balmer series) — emission from ionized gas (H II regions).
 - $H\beta$ (4861.3 Å): transition $n = 4 \rightarrow n = 2$.
 - [OIII] (5006.8 Å): forbidden emission of doubly ionized oxygen (O^{2+}) in low-density gas.
 - [OII] (3727 Å): O^+ doublet.
 - Ca II H and K (3968.5 / 3933.7 Å): absorption lines of neutral calcium.
 - Na I D (5890 / 5896 Å): neutral sodium absorption.

Redshift of galaxies

- Plot the data (spectrum for each galaxy): flux (y-axis) vs wavelength (x-axis).
- Identify the emission peaks of each galaxy, the maximum observed λ_{obs} that are identified with a line in the plot for each tracer from data measurements.
- Match the emission peaks λ_{obs} with standard spectral lines λ_0 :
 - $H\alpha \rightarrow \lambda_0 = 6563 \text{ Å}$
 - $H\beta \rightarrow \lambda_0 = 4861 \text{ Å}$
 - [OIII] $\rightarrow \lambda_0 = 5007 \text{ Å}$
 - [NII] $\rightarrow \lambda_0 = 6583 \text{ Å}$
 - Ca II K $\rightarrow \lambda_0 = 3933 \text{ Å}$
 - Ca II H $\rightarrow \lambda_0 = 3968 \text{ Å}$
- Verify that the observed galaxy lines are shifted with respect to the laboratory wavelength λ_0 . Distant galaxies show the same spectral pattern as laboratory atoms, but shifted toward redder wavelengths

$$\lambda_{obs} > \lambda_0$$

- Compute the redshift for each galaxy:

$$z = \frac{\lambda_{obs} - \lambda_0}{\lambda_0} \quad (164)$$

- Compare the computed redshift with the one reported in the dataset/catalog.
- Compute the non-relativistic velocity of each galaxy:

$$v = cz \quad (165)$$

- Compute the relativistic velocity:

$$v = c \frac{(1+z)^2 - 1}{(1+z)^2 + 1} \quad (166)$$

and equivalently:

$$1+z = \sqrt{\frac{1+v/c}{1-v/c}} \quad (167)$$

- Plot the non-relativistic and relativistic velocities (y-axis) vs redshift (x-axis).
- Verify the difference between the two velocities: For small redshift ($z < 0.1$) the two are similar, and the non-relativistic approximation can be used or makes it possible to recover Hubble's law,

$$v \approx H_0 d$$

For larger redshift ($z > 0.1$) the two velocities diverge, and the relativistic formula must be used. (The reason is explored in Hubble law activity.)

3.2 Hubble law

[25, 26, 29, 30, 11, 10, 27] **Objectives:**

- Review optics, flux, luminosity, redshift,
- Compute and plot non-relativistic velocity vs redshift or luminosity distance for each Supernovae Ia data,
- Find the discrepancy at larger redshift and introduce the Hubble law,
- Add the Hubble curve and find the best value of the slope H_0 ,
- Understand galaxy recession with distance and its limits,
- Add the relativistic velocity curve and find the discrepancy with the Hubble model,

- Plot the magnitude or distance modulus vs redshift from data,
- Add the Hubble curve (compute magnitude using Hubble distance) and compare with data,
- Introduce the concept of cosmic expansion and dark energy,
- Add the λ CDM curve (model with dark energy) and the curve only-matter (without dark energy),
- Find which is the best model that match the data.

Dataset: Pantheon-SH0ES — Type Ia supernovae data [22]: redshift, apparent magnitude, mag errors.

Procedure:

- Supernovae Ia are white dwarf in binary systems that during the system evolution grow matter from the companion star until they accumulate enough fuel matter to create an explosive reaction. Their luminosity during the explosion is universal because its peak height in brightness is equal for all the supernovae Ia. They can be observed from a long distance because of their extremely high brightness and they are called standard candles. Their distance can be estimated from their flux or apparent magnitude.
- Compute velocity for each data point (non-relativistic):

$$v = cz \quad (\text{use } z_{\text{CMB}} \text{ from the dataset}) \quad (168)$$

- The flux in cosmology:

$$f = \frac{L}{4\pi d_L^2} \quad (169)$$

depends on the luminosity distance d_L .

- Distance modulus:

$$\mu = m_B - M_B = 5 \log(d_L) - 5 \quad (170)$$

with absolute magnitude $M_B = -19.3$ and m_B the apparent magnitude from the dataset.

- Luminosity distance in Mpc:

$$d_L = 10^{(\mu+5)/5} \quad (171)$$

- Plot luminosity distance or redshift (x-axis) vs velocity (y-axis).
- Filter low-redshift supernovae and perform a linear fit to the points to find a reference straight line for the Hubble model.
- The slope that minimizes vertical distances yields an estimate of H_0 that represents the slope of the line:

$$H_0 = \frac{\sum v_i d_{L,i}}{\sum d_{L,i}^2} \approx 64 - 70 \text{ km s}^{-1} \text{ Mpc}^{-1} \quad (172)$$

- Hubble Law:

$$v = H_0 d_L \quad \Rightarrow \quad cz = H_0 d_L \quad (173)$$

(valid approximately for $z < 0.1$).

- Residuals (vertical distance of points from the line):

$$r = v - H_0 d_L \quad (174)$$

If $r = 0$ the point lies on the line; $r > 0$ indicates a galaxy receding faster than the Hubble flow; $r < 0$ indicates slower recession.

- Observation:

- For $z < 0.1$ points lie approximately on a straight line with roughly constant slope.
- For $z > 0.1$ points deviate from a single straight line: the slope changes with z .

- Compare with relativistic velocity (useful for $z > 0.1$):

$$v = c \frac{(1+z)^2 - 1}{(1+z)^2 + 1} \quad (175)$$

- Compute percentage difference between non-relativistic and relativistic velocities for each supernova:

$$\text{percentage difference} = \frac{|v_{nr} - v_r|}{v_r} \times 100\% \quad (176)$$

- In classical physics if an object moves away with velocity v , its distance after a time t is simply

$$d = vt$$

.The classical relation works only if space is static.

- However, in cosmology there is no centre from which all galaxies are being “shot out”, all observers, in all galaxies, see the same proportionality and the relation works in every direction of the sky. The galaxies are not moving through space, but it is the space itself that expands: the distance between two points changes even if the points themselves are not moving through space. Therefore, the usual formula $d = vt$ no longer works.
- If space stretches, then the distance between two galaxies at time t_1 is not the same as at time t_0 . Indeed light travels across many intervals of space, each of which expands at a different rate in cosmic history. The only correct way is to add up the contribution of every little stretch of space along the path of light, that mathematically is expressed by an integral.
- For small redshift ($z \ll 1$), we may still write the Hubble law:

$$v \approx H_0 d \rightarrow d \approx \frac{cz}{H_0} \quad (177)$$

Here the redshift z measures how much the wavelength of light has been stretched by cosmic expansion.

- Considering that H_0 is the value of the Hubble parameter at present time. At higher redshift, however, H is not constant. The expansion rate of the Universe has changed during cosmic history due to matter, radiation, curvature and dark energy. Thus, using $d = cz/H_0$ becomes inconsistent.
- For larger redshift ($z > 0.2 - 0.3$), the linear Hubble model breaks down because one must account for the stretching (dilation) of space itself: the expansion factor $a(t)$ is not constant and cosmic acceleration becomes relevant.
- Relation between redshift and scale factor:

$$1 + z = \frac{a(t_0)}{a(t_{emit})}, \quad \text{with } a(t_0) = 1 \Rightarrow a(t) = \frac{1}{1 + z} \quad (178)$$

- Relation between Hubble constant and the expansion factor:

$$v = \frac{dr}{dt} = \dot{a} r(t_0) = \frac{\dot{a}}{a} r(t) \quad \Rightarrow \quad H_0 = \frac{\dot{a}}{a} \quad (179)$$

- Hubble time:

$$t_0 = \frac{r}{v} = \frac{r}{H_0 r} = \frac{1}{H_0} \approx 14.38 \text{ Gyr} \quad (180)$$

- To handle an expanding Universe, the *comoving distance* χ is a distance measured in a coordinate system that “expands with the Universe”. In this coordinate system, galaxies that participate in the Hubble flow remain at fixed coordinates. The comoving distance does not change as the universe expands.
- The physical (or “proper”) distance at time t is:

$$D(t) = a(t) \chi \quad (181)$$

In General Relativity ,the FLRW space-time metric considers an isotropic and homogeneous universe following the cosmological principle :

$$ds^2 = dt^2 - a(t)^2 \left[\frac{dr^2}{1 - kr^2} + r^2(d\theta^2 + \sin^2\theta d\phi^2) \right] \quad (182)$$

where k is the space-time curvature, ds is the infinitesimal distance, t the time coordinate, dr is the comoving distance, r the spherical coordinates and θ, ϕ the angles. Considering a flat universe $k = 0$ and a light ray that travels to us (the angles are negligible), the equation yields:

$$ds^2 = c^2 dt^2 - a(t)^2 dr^2 \quad (183)$$

How long is the light travel from a Supernovae to us? Light travels along paths with constant velocity, so the space-time interval is null, $ds^2 = 0$ in the FLRW metric.

$$c^2 dt^2 = a(t)^2 dr^2 \rightarrow c dt = a(t) dr \rightarrow dr = \frac{c dt}{a(t)} \quad (184)$$

When a photon travels from a distant galaxy to us, it starts at time t_{emit} and arrive at time t_0 crossing regions of the Universe that expand at different speeds. Therefore, during the travel the expansion factor at the denominator is continuously changing. Replacing dr with $d\chi$, a small interval of comoving distance $d\chi$ corresponds to a physical stretching $a(t) d\chi$, giving the relation:

$$c dt = a(t) d\chi \quad (185)$$

To find the total comoving separation χ between emission and observation, all the tiny pieces $d\chi$ must add up every time (integral sum). Solving for $d\chi$ and integrating from the time of emission to the time of observation:

$$\chi = \int_{t_{emit}}^{t_0} \frac{c dt}{a(t)} \quad (186)$$

The distance is an integral because the expansion rate changes continuously over time and sum all the contributions of the expansion. The temporal distance is measured in redshift, so the integral must be converted from dt to dz . Using the fundamental redshift and expansion relation:

$$1 + z = \frac{1}{a(t)} \rightarrow a(t) = \frac{1}{1 + z} \quad (187)$$

Let $H(t)$ (Hubble parameter), be the expansion rate of the Universe at time t :

$$H(t) = \frac{\dot{a}}{a} = \frac{1}{a} \frac{da}{dt} \quad (188)$$

Applying the derivative to find the relation between dt and dz and considering $a = (1 + z)^{-1}$, differentiating to z :

$$\frac{da}{dz} = -1(1 + z)^{-2} = -\frac{1}{(1 + z)^2} \rightarrow da = -\frac{dz}{(1 + z)^2} \quad (189)$$

Applying the definition $H = \frac{1}{a} \frac{da}{dt}$ to find the time:

$$dt = \frac{da}{aH(t)} \quad (190)$$

Replacing $a = \frac{1}{1+z}$ and $da = -\frac{dz}{(1+z)^2}$ in the time equation above:

$$dt = \frac{-\frac{dz}{(1+z)^2}}{\frac{1}{1+z}H(z)} \rightarrow dt = -\frac{dz}{(1+z)H(z)} \rightarrow \frac{dt}{dz} = -\frac{1}{(1+z)H(z)} \quad (191)$$

The minus sign identifies the time is going forward ($dt > 0$) and in the past the redshift is higher (the early universe corresponds to higher redshift and it is proportional to the distance). Integrating from 0 to z instead of $z - 0$, the minus sign can be omitted. Replacing dt and $1 + z$ in $\chi = \int_{t_{\text{emit}}}^{t_0} \frac{c}{a(t)} dt$:

$$\chi = \int_0^z c(1+z) \frac{dz}{(1+z)H(z)} \rightarrow \chi = c \int_0^z \frac{dz'}{H(z')} \quad (192)$$

Considering that $H(z)$ is not constant and that in the Λ CDM model is replaced by $H(z) = H_0 E(z)$: Substituting into the previous expression yields:

$$\chi(z) = \frac{c}{H_0} \int_0^z \frac{dz'}{E(z')} \quad (193)$$

where the dimensionless expansion rate is:

$$E(z) = \frac{H(z)}{H_0} = \sqrt{\Omega_r(1+z)^4 + \Omega_m(1+z)^3 + \Omega_k(1+z)^2 + \Omega_\Lambda} \quad (194)$$

- In an expanding Universe, the luminosity distance differs from the physical distance for two reasons: photons are stretched (redshifted) by a factor $(1+z)$ (the energy is reduced by the same factor); the rate at which photons arrive is also slowed by $(1+z)$ (time dilation). The combined effect is a factor $(1+z)^2$ in the flux.
- The flux expression with the scale factor (redshift factor) included:

$$F = \frac{L}{4\pi d_L^2} \rightarrow F = \frac{L}{4\pi \chi^2 (1+z)^2} \quad (195)$$

Comparing the classical flux relation with the cosmological flux:

$$d_L^2 = \chi^2 (1+z)^2 \rightarrow d_L = \chi (1+z) \quad (196)$$

Replacing $\chi^2 = \frac{d_L^2}{(1+z)^2}$ in the flux equation found before:

$$F(\lambda_{\text{obs}}) = \frac{1}{1+z} \frac{L(\lambda_{\text{emit}})}{4\pi d_L^2} \quad (197)$$

Replacing the comoving distance integral $\chi(z) = \frac{c}{H_0} \int_0^z \frac{dz'}{E(z')}$ in $d_L = \chi(1+z)$, the luminosity distance must be larger by a factor $(1+z)$ relative to the comoving distance.:

$$d_L(z) = (1+z) \frac{c}{H_0} \int_0^z \frac{dz'}{E(z')} \quad (198)$$

- This expression gives the full expression of the cosmological luminosity distance used in the Λ CDM model providing the explanation for the accelerated expansion. The integral appears because the Universe expands at different rates at different epochs; therefore, the path of light

must be computed by summing (the integral) all contributions of the expansion history from the source to us. Replacing $E(z)$, yields:

$$d_L(z) = (1+z) \frac{c}{H_0} \int_0^z \frac{dz'}{\sqrt{\Omega_r(1+z)^4 + \Omega_m(1+z)^3 + \Omega_k(1+z)^2 + \Omega_\Lambda}} \quad (199)$$

- Typical parameter choices: $\Omega_r = 8.53 \cdot 10^{-5}$ for radiation, $\Omega_m = 0.3$ for matter (ordinary and dark matter), $\Omega_k = 0$ (no curvature or flat universe), $\Omega_\Lambda = 0.7$ for dark energy.
- In a flat Universe ($\Omega_k = 0$) and considering the radiation negligible in the present time, this leads to the relation:

$$d_L(z) = (1+z) \frac{c}{H_0} \int_0^z \frac{dz'}{\sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}} \quad (200)$$

- Plot magnitude or modulus distance vs redshift: data, Hubble relation, CDM model (only matter), Λ CDM model (dark energy).
- The distance from Hubble law: $d = \frac{cz}{H_0}$, deriving the magnitude:

$$m_B = M_B + 5 \log\left(\frac{cz}{H_0}\right) - 5 \quad (201)$$

- Reproduce the Λ CDM model on the magnitude vs redshift plot: first compute d_L using $\Omega_m = 0.3$ and $\Omega_\Lambda = 0.7$ for a flat universe ($k = 0$), radiation is negligible, then compute the apparent magnitude using the cosmological luminosity distance:

$$m_B = M_B + 5 \log\left(\frac{c(1+z)}{H_0} \int_0^z \frac{dz'}{E(z')}\right) - 5 \quad (202)$$

- Reproduce also the model with only matter $\Omega_\Lambda = 0$ and compare the curves, considering that the sum of each density Ω must be equal to one.

$$\Omega_{tot} = \Omega_m + \Omega_r + \Omega_\Lambda + \Omega_k = 1 \quad (203)$$

- If the curvature is zero (flat universe) and radiation negligible, the model with dark energy provides $\Omega_m + \Omega_\Lambda = 0.3 + 0.7 = 1$, while the model only-matter provides $\Omega_m = 1$ in order to reach the unity.
- Verify that the Λ CDM curve overlaps the supernova data. Observationally, supernovae appear dimmer than the simple linear Hubble prediction at high z , indicating accelerated expansion.
- This is the only model that reproduces the observations and it includes the dark energy component represented by the cosmological constant Λ in terms of its density parameter: $\Omega_\Lambda = 0.7$

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